Evaluation of Microgesture Recognition using a Smartwatch

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***Abstract*— Gesture based interaction and its recognition has been an area of active research with the growing popularity of wearables. We here propose an approach to detect fine-grained finger and palm motions using inertial sensors in a commercial smartwatch. A user specific SVM based classifier is developed for 7 microgestures with a classification accuracy of 94.4%. We extend this to a user adaptive model by including a few representative instances of a new user and achieve a classification accuracy of 91.7%. Further, we are able to differentiate between variations of a microgesture using three fundamental building blocks - distance, speed and orientation. A novel regression based approach is presented to predict the distance parameter. The idea is demonstrated on a swipe gesture with an error of 14%.**

Keywords—Microgestures, Continuous gestures, Smartwatch, Wearables, Controller, Touchless interaction, Machine learning.

I. INTRODUCTION

With the growing popularity of wearables and connected IoT devices, there has been an increased focus on using wearables as a controller device. Touch, voice and gestures are some of the popular modalities to interact with wearables. Interaction through gestures has currently been limited to arm movements which are detected by inertial sensors in the wearable [1]. Given that these solutions require large hand motion, the user interaction can be cumbersome and unnatural at times.

There has been recent research in detecting subtle finger and hand motions, also referred to as microgestures, using a wearable worn on the wrist. Researchers have demonstrated that microgestures can be detected by sensing the electrical field or impedance at the wrist [5,6]. But such solutions require custom sensors which are currently bulky and are not commercially viable.

In this paper we explore the possibility of detecting microgestures using existing inertial sensors in a smartwatch. In the scope of our work, we define microgestures as hand motions that require no arm movement, i.e. hand movement about the wrist and the fingers only (Fig. 1). We first evaluate the classification of such motions to establish the feasibility. Further a set of fundamental building blocks are defined that can independently or in combination define variations of microgestures.

Our main contributions in this paper are:

- Demonstrating the feasibility of using an off-the-shelf smartwatch for the detection of 7 gestures using a feature based machine learning model.
- Demonstrating a robust classification model that can successfully be trained for a new user by including a few representative instances.
- Defining and evaluating three building blocks for microgestures – distance, speed and orientation, which can be used to define variations of any gesture.
- Exploring the continuous interaction aspect of microgestures through a regression based distance prediction model.

II. RELATED WORK

There are several prior work on hand gesture recognition using sensors on wearable devices. Most of the earlier work has been in using custom sensors to detect these motions but researches have more recently been exploring the use of inertial sensors.

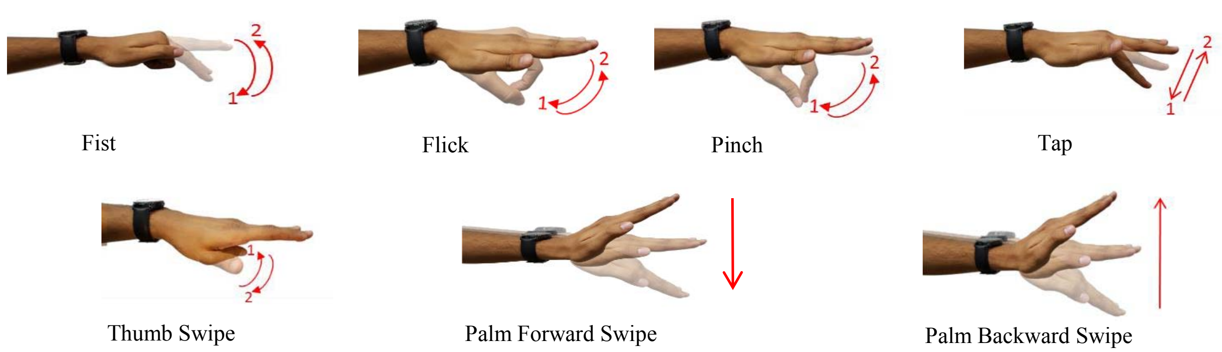

Fig. 1. Finger and palm motion based microgestures considered for the feasibility study.

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Zhang et al. have demonstrated the use of Electrical Impedance Tomography (EIT) through electrodes embedded in an armband to detect hand gestures [5,6]. Saponas et al. propose a surface Electromyography (EMG) based solution that monitors muscle signal to detect hand gestures [3]. The limitation with these solutions is that they require specialized sensors to be integrated with the wearable device. Whether the sensing is based on electric [10], magnetic field [11] or ultrasound [2], most of these methods are susceptible to one or more of the following limitations - changes in placement of the sensing device, contact condition, ambient noise, periodic removal of the device and unintentional arm movements.

The solution proposed in this paper is based on the accelerometer and gyroscope sensor in a smartwatch. Several prior researchers have demonstrated microgesture detection using inertial sensors. Xu et al. used the accelerometer and gyroscope signals from a wearable worn around the finger, wrist and forearm to detect various hand gestures performed through the movement of fingers, wrist and the arm [7]. However, their experiments had the user's wrist and arm affixed to a chair, thus limiting the generalizability of their results. ViBand captures fine-grain finger motion signals through bio-acoustics, i.e. micro vibrations captured through the accelerometer of the watch [4]. They demonstrate the detection of gestures like flick, clap, scratch, and tap, along with larger hand motions with an accuracy of 94.3% for a user specific model. Additionally, this was realized by customizing the smartwatch kernel by boosting the sensor sampling rate to 4 kHz. In comparison, our proposed approach is able to detect microgestures using existing sensor sampling rate of 100 Hz in commercial devices.

Closest to our approach is the work by Wen et al. called Serendipity [8]. It uses the accelerometer and gyroscope signal from a commercially available smartwatch to classify five fine-grain finger gestures - pinch, tap, rub, squeeze, and wave with an accuracy of 87% for 10 users. In their evaluation the training data has more than 100 instances of a test user. Though it might not be practical for a new user to provide such large number of instances before using the device.

Similar to Serendipity we have also used a Support Vector Machine (SVM) based classifier but with an extensive set of features. We have demonstrated that our proposed method has significant improvement in classification accuracy for a new user when including a few representative training instances from the user. The number of training instances is bound by the practicality of a real-world training system. We further demonstrate that the microgesture classification is independent of the hand orientation and gesture speed. In comparison, the accuracy of Serendipity drops to below 80% when gestures are performed at different speeds.

Our work also involves detection of continuous microgestures using the smartwatch. Gong et al. have developed the WristWhirl prototype that uses an array of proximity sensors to detect continuous gestures [9]. In their solution the wrist can be used as a joystick to create freeform hand movements to be recognized as gestures. In the context of the discussion in this paper, the continuous aspect of a gesture refers to detecting intermediate states of a gestures and mapping it to multi-level UI controls. We demonstrate this by predicting the distance travelled by the user's finger/palm while performing a gesture which could further be mapped to the UI.

Thus in summary, beyond the related prior works our contributions lies in proposing a holistic approach to microgesture classification. We have achieved higher classification accuracy than what is reported in the literature, even when using a commercial grade smartwatch. We further demonstrate the extension of the model to a user adaptive solution when using a limited number of representative instances from the new user.

III. METHODOLOGY

There are two key aspects to recognizing microgestures – gesture spotting and gesture classification. We here use the accelerometer and gyroscope signals from a Samsung Gear S2 smartwatch and analyze it to determine the segment of the time series corresponding to the microgesture. The segment is then classified into one of the seven gesture classes shown in Fig 1.

A. Gesture Spotting

The signal segmentation is done using the gyroscope data (g_x, g_y, g_z) as it has less noise than the acceleration signal. A threshold is set on the magnitude $\left(\sqrt{g_x{}^2 + g_y{}^2 + g_z{}^2}\right)$of the gyroscope signal to identify the start and end of the gesture. To ensure that unintended motion does not get segmented along with the gesture segment, we use the Teager Energy Operator (TEO) to dynamically alter the threshold [13].

If $x[n]$ is the magnitude of the gyroscope signal at the *n*th index, then

$$T[n] = x^2[n] - x[n-1] * x[n+1] \tag{1}$$

$$T_{th}[n] = \frac{T_{th}[n-1]*\alpha + T[n]*\beta}{(\alpha+\beta)} \tag{2}$$

where, $T[n]$ and $T_{th}[n]$ are the corresponding Teager energy and threshold respectively. α and β are constant weights assigned to the previous threshold and the current Teager energy respectively.

It is observed that the Teager threshold based approach can sometimes clip the start and end points of a gesture segment. We therefore trace the gyroscope signal backward and forward beyond the start and end point respectively to ensure that the

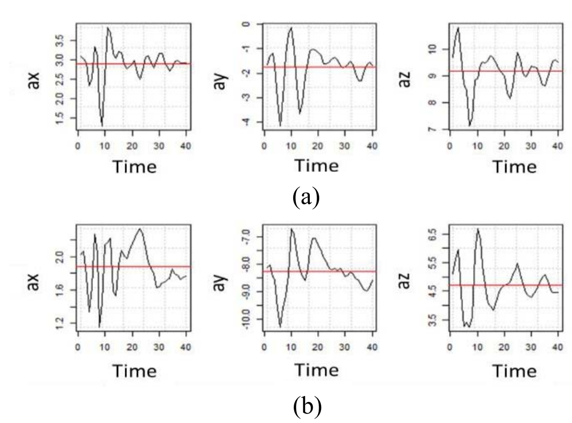

Fig 2. Accelerometer signals for Tap gesture when the palm is (a) horizontal, (b) tilted at 45°. The red line indicates the mean of the signal.

complete gesture is contained in the segment. The acceleration signal is then segmented using the updated start and end points. For the data that we have evaluated, the proposed splitting approach is successfully able to segment all gesture instances.

B. Gravity Compensation

The accelerometer of the smartwatch not only measures the change when a microgesture is performed but inherently also measures gravity. Gravity can have different components along the three axes depending on the orientation of the user's hand (Fig. 2). In order to ensure that the proposed gesture recognition system is orientation independent, we eliminate the effect of gravity by mean normalizing the acceleration segment along each axis. As shown in Fig. 2, the acceleration signal for two orientations looks fairly similar about the segment mean.

C. Gesture Classification

The classification of the segmented data is done using the SVM approach. A set of 31 features were considered, which include those suggested by Phinyomark et al. [14], Wiens et al. [15] and statistical features like root mean square, skewness, kurtosis, variance, serial correlation, zero crossing rate and power spectrums in four frequency bins. The feature values are computed for each of the gyroscope and accelerometer axes giving a total of 186 features. Given the large number of features, a feature selection approach is used to remove redundant features from the SVM model. We use Pearson correlation coefficient to remove highly correlated features.

IV. Microgesture Classification Study

The first experiment was conducted to demonstrate the feasibility of the proposed microgesture recognition system. A Samsung GearS2 smartwatch was used for data collection through a custom Tizen application that samples that accelerometer and gyroscope at 100Hz. The device was worn on the left wrist and was moderately tight. In our current implementation the gesture spotting and classification is done offline in RStudio but the logic can be easily ported onto a smartwatch for it to run in real-time.

A. Gesture Classes

A set of 7 microgestures were identified for this feasibility study - Pinch, Flick, Tap, Fist, Thumb Swipe, Palm Forward Swipe and Palm Backward Swipe as shown in Fig. 1. These microgestures are limited to movement of the finger and palm, with no movement of the arm. The inertial sensor in the smartwatch captures the subtle movement of the user's wrist as these gestures are performed.

(a) (b) (c)

Fig. 3. Various positions of hand considered for data collection - (a) unsupported in air at 0°, (b) elbow resting on table at 90°, (c) forearm resting on table at 45°.

B. Data Collection

The initial set of data was collected for a single user (User 1) with gestures performed over 10 sessions and the watch being re-worn each time. This ensured that the inherent variability in the gesture and the variability due to position of the watch on the wrist were captured in the data. Additionally, we ensured that data was collected for different positions and orientations of the hand as indicated in Fig. 3. These positions include the arm unsupported in the air, the elbow or forearm resting on the table. Additionally different orientation (0° to 90°) of the arm were considered in each of these positions.

In each session the user was asked to perform 30 instances of each gesture. A total of 300 instances were collected for each of the 7 gestures for User 1. The dataset was expanded further for 4 other users to build a user adaptive model. User 2 provided data over 5 sessions and Users 3-5 provided data over 2 sessions. In each session, these users provided 30 instances each of the 5 finger gestures - Pinch, Flick, Tap, Fist and Thumb Swipe, with different positions of the arm in each session.

C. Classification Model and Results

1) User Specific Classification Model

We use a 7-class SVM based model to classify User 1's data. A total of 110-120 features were identified after eliminating the highly correlated features from the set of 186 features. The model is trained and validated using data from 3 sessions, which corresponds to 90 instances per gesture and the remaining 7 session's data is used for testing. We use radial basis function as the kernel and obtain optimum values for cost and gamma parameter (10, 0.005) using the grid search method. In order to avoid overfitting of the model, the experiment is repeated in an 8-fold manner considering data from 3 sessions for training each time.

Fig. 4 shows the confusion matrix for the classification of 7 gestures for User 1 and indicates that there is a very high accuracy for most microgestures. The average accuracy is 94.4%. Fist gesture shows marginally lower accuracy of 89% as it gets misclassified to Pinch gesture. A similar model built for User 2, using 90 instances of each gesture in training, gave an average accuracy of 94.8%. This demonstrates, albeit for two

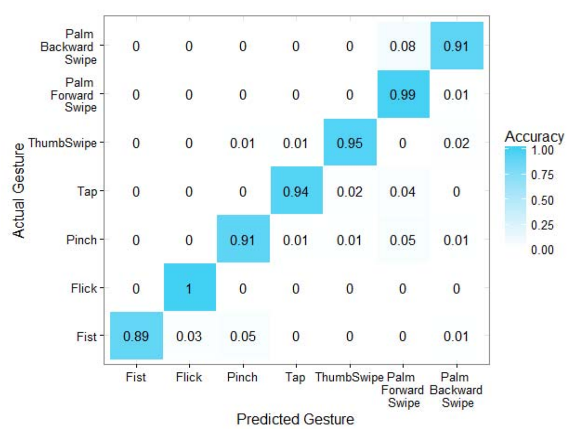

Fig. 4. Confusion matrix for the classification of 7 microgestures for User 1.

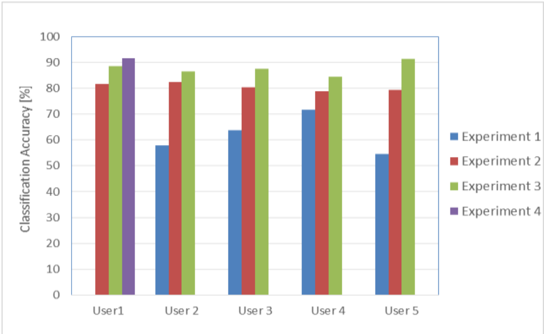

Fig. 5. Average accuracy plot for user independent study for experiments run with different training datasets.

users, that a user specific model can be built with high accuracy if there are significant representative instances in the training data. This is similar to Serendipity where they have used more than 100 instances of a new user and achieved an average accuracy of 87% [8].

Though the classification accuracy is high with a user specific model, it might be impractical as a commercial solution. It is unlikely that a new user would provide more than 100 instances upfront to train the model. With this in mind, we evaluate a user adaptive model which is within the requirements and practical limitations of a commercial solution.

2) User Adaptive Classification Model

To explore the possibility of classifying microgestures with a user independent model, we use the model designed in the previous experiment with User 1's data and test it with the data from the 4 other users (*Experiment 1*). Average classification accuracy was only 62%, suggesting that a model trained with only one user's data would not work for other users (Fig. 5). This is primarily because each user's hand structure affects the gyroscope and accelerometer signal differently. Moreover, training data from one user alone is not sufficient to capture enough variation in the gesture motion.

As a subsequent evaluation, an SVM model was built using data from 4 users as the training data (*Experiment 2*) and tested on the 5th user. Data from one session of each user was used giving a total of 120 training instances per gesture class. For a 5-fold experiment, the average classification accuracy improved to 80.6% (Fig. 5). This reaffirmed our hypothesis of the model needing sufficient gesture variations for better classification. We expect the accuracy to further improve by including instances from additional users.

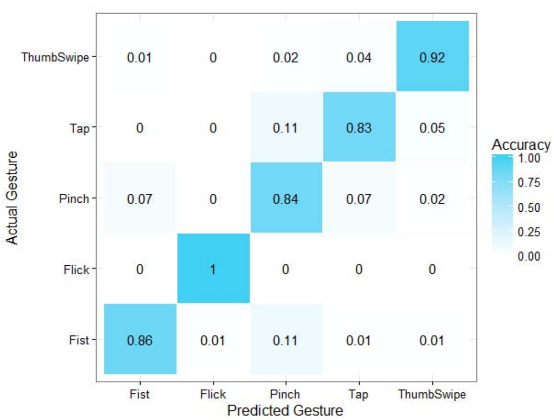

Fig. 6. Confusion matrix for the classification of 5 microgestures across 5 users, using training data as per Experiment 3.

To further improve the classification accuracy, we propose to add a few instances of the new user in the training model (*Experiment 3*). This would be a user specific model while still being within practical constraints of data collection. We have evaluated this by including 10 instances of each gesture from a given session of the test user with 120 training instances of the other 4 users. The model was tested on the user's data from other sessions so as to avoid overfitting. The average classification accuracy improved to 87.8% for a 5-fold validation (Fig. 6). Accuracy for Pinch and Tap gestures are lowest, primarily because of significant overlap between the two gestures. The motion of the index finger in the two gestures is nearly the same.

Though the average accuracy significantly improves by adding representative instances, it isn't as high as the user specific model (94% for User 1 and 2). This is because all 10 instances that were included in Experiment 3 were from the same session. Gestures performed in a single session had little variation across instances as the user's hand position and orientation were fixed. We propose that the accuracy can be further improved by including instances with enough variation. To test out the above, we considered User 1 as the test user as we have data from 10 sessions. A single instance of each gesture is taken from each of the 9 sessions and included in the training model (*Experiment 4*). Data from the 10th session is used to determine the accuracy. This experiment is repeated in a 10-fold manner and an average accuracy of 91.7% is achieved. This experiment demonstrates that the proposed model even with limited representative instances from a new user can achieve high classification accuracy. The accuracy of our model is higher than prior published work while using significantly fewer training instances.

V. Microgesture Building Blocks

A microgesture can be performed in different ways by varying one or more of the following parameters at a time – (1) Orientation, (2) Speed, and (3) Distance. We refer to these parameters as the microgesture building blocks, as each of these independently defines a unique aspect of a microgesture.

A. Distance

This parameter is a measure of the travel distance by the palm and/or the finger while performing a microgesture. Movement of both palm and individual fingers is limited by the

(a) (b)

Fig. 7. (a) Finger Vertical Swipe, (b) Finger Horizontal Swipe.

pivotal joints. A user can have sufficient control of this travel distance. For instance, the Finger Swipe gesture as shown in Fig. 7(a) can have several intermediate states between the initial and final finger position (0° to 90°). These gesture states by virtue of distance travelled can be intuitively mapped to different levels of interactions in applications for volume, brightness control, seek bar etc. In our proposed approach, we use the accelerometer and gyroscope signals to predict the distance as discussed in the following section.

B. Speed

Speed is a measure of the time taken for the palm and/or the finger to move from an initial to a final stationary state while performing a microgesture. User can consciously perform the same microgesture at faster or slower speeds to trigger different interactions in the application. This is analogues to the aspect of speed in touchscreen gestures like swipe and flick.

C. Orientation

Microgestures can also be classified based on the orientation of the finger and/or palm while performing the gesture. Possible orientations for the finger or palm of the user's hand can be facing up, down, sideways or any other intermediate orientation. For instance the Finger Swipe gesture can be performed by moving the finger up to down (Fig. 7(a)), or by moving the finger left to right (Fig. 7(b)).

In the following section we present the evaluation of our algorithm's accuracy in classifying microgestures based on one or more of these building blocks.

VI. Microgesture Building-Block Experiment

In this section we demonstrate the detectability of the microgesture building block parameters using specific gestures.

A. Distance

We here evaluate the detectability of the travel distance for the Palm Forward Swipe gesture. As the distance parameter is to be correlated to the inertial sensor signal, we use an SVM based regression model to predict the distance travelled.

The user was asked to perform the Palm Forward Swipe gesture with random start and end positions of the palm such that different travel distances were captured across the instances. The key challenge with determining the distance is the ground truth information. We used a video camera setup to monitor the hand movement as the user performed the gesture. The user was asked to wear a red colored ring while performing the gesture so that image processing techniques could be used to track the effective

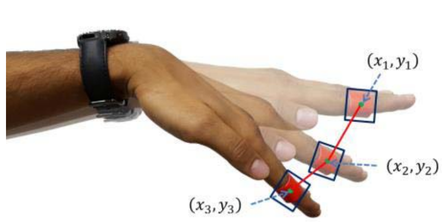

Fig. 8. Illustration of colored ring identification and distance calculation.

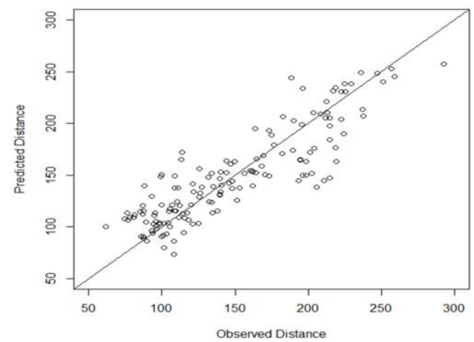

Fig. 9. Observed and predicted gesture distance in pixel units.

finger movement (Fig. 8). As we use a single camera to capture the motion, care was taken to perform the gestures at a fixed distance from the camera so that the distance travelled could be computed accurately. The captured video at 30fps is synced with the inertial sensor measurement at 100Hz. The gesture spotting approach discussed in earlier sections is used to segment the inertial sensor data and then to identify the corresponding video frames. Each frame of the video was resampled to a grid of size 512 x 380. For each frame, a rectangular area corresponding to the ring was identified using a threshold on the color red. The pixel location corresponding to the centroid of the rectangle was determined for each frame. For a given instance, the travel distance as computed from the captured video segment is the sum of the Euclidean pixel distance (in pixel units) across the frames. For the 660 instances captured during our study, this distance varied from 70 to 300 pixel units (Fig. 9).

An SVM based regression model using the gyroscope and accelerometer data is used to predict the travel distance. We used the eps-regression technique implemented in e1071 package of RStudio. The same feature set used for the classification model in the previous section is used here for regression. 510 instances of the gesture with distance measured from the vision system are used to train and validate the model and 150 instances are used to test the model. Similar to the classifier model, the grid search method was used to obtain the optimum values of epsilon, cost and gamma parameters (0.2, 32 and 0.001). It is observed that the predicted distance by the regression model for the training and test data has an average error of 6% and 14% respectively.

Fig. 10 shows the distance prediction error distribution for the 150 test instances and it is evident that the error is less than

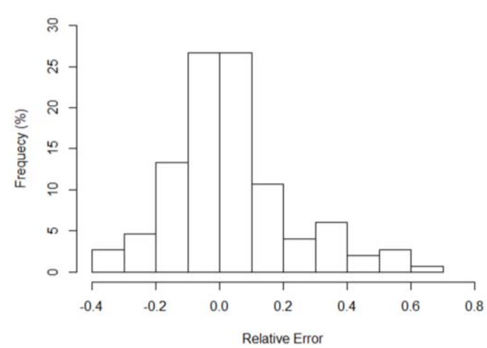

Fig. 10. Histogram of relative error for predicted vs actual gesture travel distance.

20% for 80% of the instances. Additionally, from Figure 9 the RMS of the absolute error for the test data is 25 pixel units. Though the error in the predicted distance is slightly high, it might still suffice for practical applications like volume and brightness control where the granularity of control is not necessarily continuous but limited to a few discrete levels.

B. Speed

The speed parameter for the microgestures is evaluated by determining the average speed when the gesture is performed. We use the regression model for distance prediction and the gesture time to compute the average speed. The user is asked to perform the Palm Forward Swipe gesture with varying speeds to capture a range of speeds.

We also use the SVM classifier model developed for the feasibility study experiment to confirm that the model is robust to speed variations. A classification accuracy of 98% was achieved. The few misclassified instances were not necessarily in the extreme speed ranges. Using the speed information the instances can be mapped to different behavior of the same gesture. In comparison, the accuracy of the Serendipity model drops to below 80% when gestures are performed at different speeds.

C. Orientation

In our microgesture classification experiment, instances for each gesture were collected for different orientations of the hand. To demonstrate the robustness of the classifier to hand orientation, we compensate for gravity from the accelerometer signal. Although we have designed an orientation independent model, it is equally possible to design an orientation dependent model. The absolute orientation of the hand can be easily determined from the measurement of components of gravity along each axis, which can be used to interpret the corresponding gesture differently. For example, the Palm Forward Swipe gesture can be interpreted differently when performed horizontally or vertically. The gesture can be mapped to a horizontal or vertical scroll when controlling a UI.

VII. Conclusion And Future Work

We present the evaluation of a microgesture recognition model that detects finger and palm based gestures using existing inertial sensors in a smartwatch. A 7-class SVM model was designed to classify microgestures, including Pinch, Flick, Tap, Fist, Thumb Swipe, Palm Forward Swipe and Palm Backward Swipe. Results show the average classification accuracy to be 94.4%, which is significantly higher than related prior works. We further extend the model to be user adaptive by including only 10 instances from a new user. We then achieve a classification accuracy of 91.7% when the chosen instances capture the variability due to different hand positions.

We further demonstrate that our system can accurately distinguish between variations of microgestures based on three proposed building blocks - distance, speed, and orientation. The prediction of the distance parameter is demonstrated on the Palm Forward Swipe microgesture. The proposed SVM based regression model uses the accelerometer and gyroscope data and has an average prediction error of 14%.

We believe that our work has contributed in important ground work for exploring the potential of smartwatches to be an interaction device of the future. Our next step would be to assess the accuracy of the proposed model across more users and further explore the design of a completely user independent model. The robustness of the solution would also have to be evaluated based on its ability to reject non-gestures.

References

[1] Sonu Agarwal, Arindam Mondal, Gurdeepak Joshi and Gaurav Gupta. 2017. GestGlove: a wearable device with gesture based touchless interaction. In *Proceedings of the 8th Augmented Human International Conference (AH'17)*.

[2] SoundSense: 3D Gesture Sensing using Ultrasound on Mobile Devices.2017. Retrieved April 3, 2017 from http://mrorz.github.io/files/soundsense.pdf

[3] Scott. T. Saponas, Tan. S. Desney, Dan Morris, Ravin Balakrishnan, Jim Turner, and James A. Landay. 2009. Enabling always-available input with muscle-computer interfaces. In *Proceedings of the 22nd annual ACM symposium on User interface software and technology (UIST'09)*, 167-176.

[4] Gierad Laput, Robert Xiao, and Chris Harrison. 2016. ViBand: High-Fidelity Bio-Acoustic Sensing Using Commodity Smartwatch Accelerometers. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST'16)*, 321-333.

[5] Zhang Yang, and Chris Harrison. 2015. Tomo: Wearable, low-cost electrical impedance tomography for hand gesture recognition. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (UIST'15)*, 167-173.

[6] Zhang Yang, Robert Xiao, and Chris Harrison. 2016. Advancing Hand Gesture Recognition with High Resolution Electrical Impedance Tomography. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST'16),* 843-850.

[7] Chao Xu, Parth H. Pathak, and Prasant Mohapatra. 2015. Finger-writing with smartwatch: A case for finger and hand gesture recognition using smartwatch. In *Proceedings of the 16th International Workshop on Mobile Computing Systems and Applications*, 9-14.

[8] Wen Hongyi, Julian Ramos Rojas, and Anind K. Dey. 2016. Serendipity: Finger gesture recognition using an off-the-shelf smartwatch. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '16), 3847-3851.

[9] Jun Gong, Xing-Dong Yang, and Pourang Irani. 2016. WristWhirl: One-handed Continuous Smartwatch Input using Wrist Gestures. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST'16)*, 861-872.

[10] Junhan Zhou, Zhang Yang, Gierad Laput, and Chris Harrison. 2016. AuraSense: Enabling Expressive Around-Smartwatch Interactions with Electric Field Sensing. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST'16)*, 81-86.

[11] Chris Harrison, and Scott E. Hudson. 2009. Abracadabra: wireless, high-precision, and unpowered finger input for very small mobile devices. In *Proceedings of the 22nd annual ACM symposium on User interface software and technology (UIST'09)*, 121-124.

[12] Rajalakshmi Nandakumar, Vikram Iyer, Desney Tan, and Shyamnath Gollakota. 2016. Fingerio: Using active sonar for fine-grained finger tracking. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '16), 1515-1525.

[13] Kaiser, J.F., 1990, April. On a simple algorithm to calculate the energy of a signal. In *Acoustics, Speech, and Signal Processing, 1990. ICASSP-90., 1990 International Conference on* (pp. 381-384). IEEE.

[14] Phinyomark, A., Nuidod, A., Phukpattaranont, P. and Limsakul, C., 2012. Feature extraction and reduction of wavelet transform coefficients for EMG pattern classification. *Elektronika ir Elektrotechnika*, *122*(6), pp.27-32.

[15] Wiens, J., Horvitz, E. and Guttag, J.V., 2012. Patient risk stratification for hospital-associated c. diff as a time-series classification task. In *Advances in Neural Information Processing Systems* (pp. 467-475).